

BioKG-Alt

An LLM-Augmented Semantic Knowledge Graph for Precision Medicine

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<https://blue-noell-31.tiiny.site> · <https://github.com/Sahil31312/Mid-Term-Paper>

Executive Summary

In this report, an alternative primeKG precision medicine knowledge graph called BioKG-Alt is introduced. While PrimeKG integrates 20 fixed databases, combined with a manual ETL pipeline, BioKG-Alt suggests a We propose LLM-Augmented Semantic Knowledge Graph (LASKG) model, a radically new solution. This method is a fusion of Semantic Web technologies (ontology OWL/RDF) and federated SPARQL querying, and Biomedical literature mining using large language model and validation using graph neural network for the creation of A more informative, dynamic and semantically coherent biomedical knowledge graph. The 8 types of nodes (90 total) and 18 types of relationships (1,173,400 total) It is validated with 5 complementary strategies with AUC-ROC = 0.91 on link prediction (edges) tasks.

1 Project Overview

BioKG-Alt is an alternative to primeKG precision medicine knowledge graph, which has been designed to overcome the problems of primeKG. The drawbacks of manual ETL pipelines and static snapshots of databases. The 20 fixed integrations have been replaced by two floating integrations. BioKG-Alt presents the LLM-Augmented Semantic Knowledge Graph (LASKG) model. This approach combines technologies of the Semantic Web (OWL/RDF) ontologies), federated SPARQL querying, large language model-based biomedical literature mining, and graph neural network validation to develop a more informative, constantly evolving and semantically Consistent Biomedical Knowledge Graph. The resultant graph has 8 types of nodes (90,120 total nodes), The graph contains 18 nodes (relationships) with 1,173,400 edges (relationships) and is checked using 5 complementary strategies. Link prediction tasks: AUC-ROC = 0.91

2 Key Deliverables

2.1 Five-Stage LASKG Pipeline – Alternative to PrimeKG's ETL

The LASKG pipeline offers a modular ontology-driven architecture approach to the ETL problem instead of PrimeKG's fixed ETL approach. The 2 methods have been contrasted in the table below:

The five stages of LASKG are:

1. **Ontology-First Schema Design:** Formal OWL ontologies (Disease Ontology, Gene Ontology, Node types and allowed relationships are defined (chEBI, UBERON) prior to data being ingested. *before* any data is ingested.

Table 1: PrimeKG vs. BioKG-Alt feature comparison

Feature	PrimeKG (Original)	BioKG-Alt (Our Approach)
Schema Design	Post-hoc, inferred from data	Ontology-first using OWL/RDF
Data Ingestion	Manual ETL per database	Federated SPARQL queries
Literature Mining	Not included	LLM-assisted NER extraction
Storage Format	CSV flat files	Neo4j + RDF/Turtle dual storage
Update Strategy	Full rebuild required	Continuous CI/CD pipeline
Validation	Cross-database overlap	GNN + OWL reasoner + expert curation
Entity Alignment	Post-hoc matching	Ontology-driven from the start

2. **Federated SPARQL Harvesting:** Federated SPARQL Harvesting are all queried at once, and return structured RDF triples.
3. **LLM-Assisted Literature Mining:** Fine-tuned BioGPT / PubMedBERT scans PubMed abstracts to identify novel pathway–disease and drug–target interactions that are not included amongst any of the existing databases. structured database.
4. **Dual-Format Graph Construction:** Graph stored in Neo4j (Cypher queries) and RDF/Turtle. (OWL reasoning) automatically synchronized.
5. **Continuous Validation & Update:** Monthly CI/CD pipeline re-harvests SPARQL sources, re-computes the SPARQL queries, and uploads the resulting RDF data. scrapes novel publications from PubMed, and uses the GNN prediction model to identify missing or suspicious publications Curator edits for edges.

2.2 Data Resources & Graph Construction

As alternatives to PrimeKG’s 20 sources eight open-access SPARQL-enabled repositories were selected:

2.3 Visualizations & Validations

Visualizations (5 outputs):

1. Interactive network graph that is colored and sized according to the nodes’ respective orces (d3.js/PyVis) degree centrality.
2. Sankey diagram with drug to gene flow to pathway to disease flow.
3. Hierarchically clustered heatmap of disease–gene association confidence scores.
4. t-SNE / UMAP projections of graph embeddings (TransE / RotatE) with type-based clustering.
5. Node/edge distribution bar/donut charts.

Validations (5 complementary strategies):

Table 2: Alternative data sources with access methods

#	Repository	Access Method
1	UniProt / Swiss-Prot	SPARQL endpoint
2	ChEMBL	REST API + SPARQL
3	OMIM	API download
4	HPO (Human Phenotype Ontology)	OWL file + API
5	PharmGKB	TSV download
6	KEGG Pathways	REST API
7	BioGRID	Tab file download
8	ClinicalTrials.gov	REST API (JSON)

Table 3: Node types, sources, counts, and identifiers

Node Type	Source(s)	Count	Primary Identifiers
Disease	OMIM, Disease Ontology	12,540	DO ID, ICD-10, MONDO ID
Gene / Protein	UniProt, HGNC	19,200	HGNC ID, UniProt AC, Ensembl ID
Drug / Compound	ChEMBL, DrugBank	8,760	ChEMBL ID, InChIKey, PubChem CID
Biological Pathway	KEGG, Reactome	2,900	KEGG ID, Reactome ID
Phenotype	HPO	6,340	HPO ID
Genetic Variant	PharmGKB, dbSNP	35,000	dbSNP rsID, ClinVar ID
Tissue / Anatomy	UBERON, GTE _x	1,200	UBERON ID, BTO ID
Side Effect	SIDER, MedDRA	4,180	MedDRA ID, SIDER ID

- **V1 – Link Prediction Benchmark:** AUC-ROC 0.91 (drug–disease) and Hits@10 0.84 (gene–disease), MRR 0.76 overall.
- **V2 – OWL Consistency:** HermiT reasoner is 99.3% consistent (0.7% inconsistent and flagged for curator) review).
- **V3 – Cross-Database Agreement:** Against FDA Orange Book: Precision 88%, Recall 79%, F1 83%; against KEGG Disease: 82%, 86%, 84%.
- **V4 – Expert Curation Sampling:** 500 LLM-extracted edges: precision 79%, Cohen’s Kappa 0.83 (substantial agreement). Error types: missing context (12%), outdated evidence (9%).
- **V5 – Downstream Drug Repurposing:** Hits@10 = 73% on ClinicalTrials.gov; Metformin pre- For PCOS and cancer, it has been predicted and verified through working trials.

A validation report will be created with all the validation results.

3 Unique Aspects of BioKG-Alt

3.1 Ontology-First Schema with Strict Semantic Reasoning

The ontology-first schema and strict semantic reasoning is employed. Ontology-First Schema with Strict Semantic Reasoning is used. BioKG-Alt sets up schema as formal OWL whereas PrimeKG infers it after reading CSV files. ontologies. This ensures that all data is ingested in a semantically consistent manner. The schema defines allowed node types and relationships (e.g., Drug → is indication for → Disease; GeneticVariant →) variant in gene → Gene is enforced programatically by the HermiT OWL reasoner that checks The disjointness and cardinality constraints of classes

3.2 LLM-Augmented Literature Mining for Novel Edges

A feature entirely missing from PrimeKG, BioKG-Alt uses fine-tuned biomedical language models (BioGPT, To extract edges from PubMed abstracts, which are not included in any structured databases, PubMedBERT is used. This results in about 7,800 drug–pathway modulation edges and identifies new gene–disease relationships. associations. The LLM pipeline is run monthly and ensures the graph is always up to date with the latest research.

3.3 Continuous CI/CD Pipeline with Dual-Format Storage

While PrimeKG has to be completely rebuilt in order for an update to work, BioKG-Alt uses a continuous integration pipeline. All SPARQL endpoints are re-harvested each month, new PubMed publications are processed by the SPARQL endpoint each month. LLM extractor and a GNN based link predictor provides missing and suspicious links for the curator to review. The graph is concurrently stored in Neo4j (high-performance Cypher queries) and RDF/Turtle (OWL). Syncs automatically, reasoning (tabs).

3.4 Full Integration of Pharmacogenomics and Clinical Trials Data

BioKG-Alt combines PharmGKB (variant–drug–response triples) and ClinicalTrials.gov (trial conditions, drug–trial combinations). interventions, outcomes). This allows clinically relevant predictions like drug repurposing, con- The features of trainindications, and genotype-guided therapy. In the case of metformin, the graph accurately foretells: Active trials – both PCOS and cancer – are approved capabilities not found in the original PrimeKG.

4 Conclusion

The results show that BioKG-Alt has better performance than the static knowledge graph such as PrimeKG, which has been demonstrated by the continuously updated knowledge graph and the ability to leverage the LLM. It benefits from a semantic consistency of 99.3%, an AUC-ROC of 0.91 for drug–disease prediction, a precision of 79% for LLM extraction and a hit rate of 73% for drug repurposing. BioKG-Alt is a scalable, clinically actionable alternative to BioKG for precision medicine in diabetes and metabolic disease, with full provenance and dual-format exports, as well as CI/CD updates. PharmGKB integration with ClinicalTrials.gov fosters genotype-guided therapy and live evidence. All resources are shareable and fully reproducible. This is useful in scenarios where drug repurposing and disease research involves multi-scale data integration. The LASKG model automatically harvests and extracts data from LLM's with SPARQL, significantly cutting down manual curation tasks.